TABLE OF CONTENTS

Description - This section contains theory for the magnet enclosure-related hardware. Sheet numbers listed at the end of some subsection titles refer to the sheet number within the Patient Handling Block Diagrams.

1- OVERVIEW

Hardware covered in this section includes the Scan Room Interface (SRI) Module, Magnet Enclosure Displays, Dock Assembly, Physiological Acquisition Module (PAC), and Magnet Bore Temperature Sensor for Signa Horizon systems.

2- SCAN ROOM INTERFACE (SRI) MODULE (MG2A33)

The SRI Module, which houses the SRI board (MG2A33A1), is an RF-tight enclosure mounted on the magnet enclosure. The RF-tight enclosure is required to prevent generation of RF artifacts from the microcontroller, and to prevent degradation of the microcontroller operation due to RF and gradient pulses.

The SRI incorporates an 80C196 microcontroller to perform its functions, which include:

- Monitoring operator control / display boards and reporting switch state changes to the TYME board (MR2A15A23) located in the TPS.
- Driving longitudinal position and scan time remaining displays on operator control/display boards.
- Controlling cradle motion.

The 80C196 is controlled by a program stored in EPROM (U140) and SRAM (U145). Jumper J5 on the SRI board can be installed to make 8K of EPROM available for stand-alone diagnostics. When J5 is not installed, the same 8K is available in SRAM instead of in EPROM.

The SRI receives +24 Vdc, +12 Vdc, and +8 Vdc power from three power supplies located on the Penetration Cabinet Power Supply Module. The 24 Vdc power supplies current to switches on the operator control / display boards. The 12 Vdc powers the seven-segment displays for longitudinal position and scan time remaining. The 12 Vdc is regulated to provide +8 Vdc for the dock and home limit switch boards (MG2A31A1 and MG3A1A1, respectively). The 12 Vdc is also regulated, via an adjustable voltage regulator, to 5.2 Vdc to power the Longitudinal Encoder (MG3A6). The 8 Vdc is regulated to power the +5 Vdc logic circuitry on the SRI board. A power-up reset circuit suspends microcontroller operation until the 5 Vdc power is stable.

The 80C196 is put into low power "sleep" mode during scanning to minimize EMI emissions. During "sleep" mode, all external bus activity is suspended until the 80C196 receives an interrupt to "awaken" it. Any switch closure produces an interrupt to "awaken" the 80C196. The SRI takes the 24 Vdc signals from the Control Switch boards (MG2A4 and MG2A5) and, through voltage dividers, provides TTL-compatible level for storage in memory. The 80C196 communicates switch closure status via fiber optic link to the system cabinet TYME board (MR2A15A23) and then over the system cabinet backplane to the IPG board (MR2A15A20).

2- SCAN ROOM INTERFACE (SRI) MODULE (MG2A33) (continued)

Devices TR207 and TR208 provide fiber optic receive and transmit portions of a full duplex serial link to the IPG board. Device TR243 is a fiber optic receiver that monitors the remote reset line from the IPG. The transmitter (TR208) is powered from the 12 Vdc supply through a current-limiting resistor. The fiber optic devices transmit and receive light in the infrared region of the electromagnetic spectrum.

The SRI controls cradle motion (speed and direction into and out of magnet bore) by providing a +10V to –10V analog signal (MOTORCMD and MOTORRTN) to the Longitudinal Drive Power Amplifier board (PP2A17A2), which powers the longitudinal drive motor. As a safety feature, the SRI generates and sends a *cradle motion enable* signal to the Manifold Interface Board (PP2A17A1). The cradle motion enable signal must be present for cradle motion to occur. If the SRI fails, this signal is not generated and cradle motion is disabled. The SRI determines cradle position by input from the Longitudinal Opto Encoder (MG3A6), and displays this position directly. One exception is landmark, at which position the SRI is directed by the IPG to zero out the longitudinal position display.

3- MAGNET ENCLOSURE DISPLAYS

Signals related to the Longitudinal Position Display, Scan Time Remaining Display, "S" and "I" indicators, and two illuminated switches, are processed by the IPG located in the TPS chassis of the system cabinet (MR2). Packets of information related to the display are sent from the IPG to the TYME board, and then over fiber optic link to the SRI. The SRI decodes the packets, and display drivers send current to the seven-segment displays and other miscellaneous indicators on the operator control/display boards.

Display drivers are open collector devices that provide a current return path from the indicator LEDs on the operator control / display boards. Each driver output line penetrating the scan room wall has a two-section RC filter providing approximately 80 dB attenuation at 7 mHz.

4- DOCK ASSEMBLY (MG2A29)

The dock assembly provides the means to secure the patient transport to the magnet enclosure, and it controls the vertical positioning of the cradle. The dock assembly contains an electric motor that receives 30–48 Vdc power from the Dock Power Supply (PP2A14) located in the Penetration Cabinet (PP2). The Dock Motor On switch, located on the inside of the dock base, is cam-activated when the dock Up pedal is pressed. Closing the normally open switch allows current to flow through the switch to the dock motor. The dock motor drives an eccentric cam that transforms the circular motion of the gear motor into reciprocating motion. This motion is transferred to the patient transport via a push rod that actuates the hydraulic pump on the transport, supplying fluid to the hydraulic cylinder, thus raising the table.

An Up Limit switch mounted on the dock base is mechanically activated by a mechanism in the patient transport when the cradle is within 1/4-inch of its Full Up position. When the Up Limit switch is activated by an actuator in the patient transport, a signal is generated on the Dock Limit Switch board signifying a table Full Up position. A signal is then sent to the interrupt controller, which provides switch status to the SRI.

4- DOCK ASSEMBLY (MG2A29)(continued)

The dock assembly also contains a down valve push rod which is cam activated by pressing the dock Down pedal. This actuates the down valve on the hydraulic cylinder in the patient transport, causing the table to lower. A spring returns the down valve push rod to its normal position.

5- PHYSIOLOGICAL ACQUISITION CONTROLLER (PAC2)

(Refer to TPS/RF Block Diagram)

The gated studies hardware synchronizes (gates) scanning at a particular time, based on physiological information from the patient. The information can take several forms including cardiac, respiratory, or peripheral pulse activity.

Cardiac gating is based on the patient's heart motion, which is monitored by ECG (electrocardiogram) leads placed on the chest. This is used when the specific position of the heart is important for imaging. Respiratory compensation helps eliminate artifacts caused by patient breathing motion. It can be based on the respiratory rate, which is measured by an air bellows placed across the abdomen. Peripheral gating minimizes the effect of blood flow through the body. A plethysmograph (photopulse sensor) placed on the patient's finger detects blood flow.

All three of these gating techniques share the same hardware, which consists of the Physiological Acquisition Controller (PAC) module. A newer version PAC, named PAC2, provides the same type of functions as the original Signa PAC. The PAC2 board acquires, digitizes, and transmits physiological signals from the patient to the Signal Processing Unit (SPU) of the Integrated Pulse Controller and Gating (IPG) board. The system uses these signals to synchronize the scanner to physiological events, such as patient ECG, pulse, and respiration.

The PAC2 has the following features:

- Twelve-bit A/D converter for acquisition of Resp., P-Pulse, and Aux.-Controllable offset voltage adjustment for respiration
- Isolated ECG signal amplifier circuits for acquisition of ECG Lead II and Lead III (Lead I is constructed by the SPU)
- Controllable gain adjustment for ECG and respiration
- Respiration signal detection (based on air pressure)
- Peripheral pulse signal acquisition
- Two auxiliary analog inputs
- Diagnostic signal channels
- Sixteen-bit A/D converter on isolated circuit
- Bar graph LED for peripheral pulse and respiration driven by the analog signal
- An 80C196 microcontroller running at 12 mHz
- 32Kbytes of EPROM; 32 Kbytes of SRAM
- High-speed fiber optic serial port to SPU
- One UART debug port
- One spare UART channel

5-1 PAC2 CPU

The PAC2 board is controlled by an 80C196 16-bit microcontroller (CPU) operating at a frequency of 11.059 mHz. The microcontroller is responsible for controlling the A/D and D/A convertors on the board that are used to acquire various physiological signals. The CPU has access to 32 kilobytes (arranged as 16K x 16K) of EPROM and 32 kilobytes (arranged as 32K x 8K) of CMOS static RAM.

A DUART with two serial ports is also connected to the CPU. One port interfaces with a debug termina, and the other is reserved as an auxiliary serial port for future use. Both serial ports are RS232-compatible. The 80C196 CPU contains one built-in serial port, which is used for communications with the SPU on the IPG board via a high-speed serial link (345K). The PAC2 board includes a fiber optic transmitter and a receiver. Physiological data from the 80C196 CPU is transferred to the SPU via this high-speed fiber optic line.

A high-speed PAL on the PAC2 provides address decoding for the microprocessor, along with wait state generation. A MAX PAL (multiple array matrix programmable array of logic) provides support services for the PAC2.

The PAC2 uses a TL7705A voltage supply supervisor integrated circuit to properly reset the 80C196 microprocessor, and to support logic on power up and if the power supply drops below 4.5 Vdc. At power up, the IC holds the reset line low for approximately 20 msec until the power supply has stabilized. In addition, if the power supply drops below 4.5 volts, the reset signal is activated for this time period. The reset line on the 80C196 microprocessor is bidirectional. It normally operates as an input; however, if a soft reset is issued by the microprocessor, this line acts as an output. In addition, the microprocessor has a built-in watchdog timer that makes sure that the PAC2 is operating properly. If the PAC2 software fails to write to a register in a certain time period, the reset line on the 80C196 reactivates.

5-2 PAC2 A/D

The main A/D converter on the PAC2 board is a 16-bit A/D. This A/D digitizes the nonisolated waveforms. An eight-channel serial multiplexer, in front of the A/D converter, controls switching between eight channels of data.

5-3 PAC2 ECG Circuit

The entire ECG front end is floating relative to earth ground. Power is provided by an air core transformer. The only connections between the ECG front end and the rest of the PAC2 board are through opto couplers. The ECG front end acquires ECG leads II and III and digitizes. A dual 8-bit DAC adjusts the gain of each channel independently, and both channels are ac-coupled, so no offset adjustment is required. The ECG front end is also capable of detecting a lead fail or lead off condition by sensing the dc offset on the ECG leads. The ECG front end is connected to a patient by the ECG lead cable that plugs into the front of the PAC assembly. The four ECG leads pass through RF filters and go to connector J2 on the PAC2 board. The RL lead is used as a drive lead to bias the circuit such that the circuit ground is at a potential of –2.5 volts relative to the patient. Thus, the three ECG leads RA, LA, and LL have a dc offset of 2.5V relative to isolated ground. The RA and LL leads are routed to an instrumentation amplifier with a fixed gain of 100x to produce ECG lead vector II. Similarly, LA and LL are routed to an instrumentation amplifier with a fixed gain of 100x to produce ECG lead vector III.

5-3 PAC2 ECG Circuit (continued)

After the instrumentation amplifiers, ECGII and ECGIII are ac-coupled and low pass filtered to remove unwanted offsets, and to prevent aliasing by the A/D converter. The signals are then sent to a dual D/A converter configured as a multiplier with a gain in the range of 1 to 256. The gain is programmed by the 80C196 via interface logic in the PAC2 control logic section. The lower the value written to the DAC, the higher the gain. After the gain stage, the signals are level shifted so as to be compatible with the LTC1290 A/D converter, which is in a unipolar configuration. After the level shifting, the signals pass through a fixed gain stage of 4 and a clipping network before going to the A/D converter. At this point, the signals are digitized and sent via opto couplers to the microprocessor interface logic. In addition to going to the instrumentation amplifiers, the three ECG leads RA, LA, and LL are routed to three noninverting buffers. The output of the buffers are low-pass filtered to prevent aliasing, and buffered again before being sent to the LTC1290 A/D converter for digitizing. The microprocessor uses these three inputs, together with RL, to detect a lead fail condition.

A CS5102A 16-bit serial A/D converter digitizes the ECG waveforms. The A/D is programmed by writing a control byte to the A/D. A 16-bit data word may then be read back from the same location.

Unlike the original PAC which used a piezoelectric transformer, the PAC2 uses an air core transformer as an isolated power supply for the ECG circuitry.

5-4 PAC2 Respiratory Circuit

A rubber bellows, wrapped around the chest of a patient, produces an air pressure signal that is proportional to the patient's breathing pattern. The bellows is connected to the front of the PAC2 assembly through a locking, pressure-tight connector. A PVC tube is routed from this connector to an air pressure transducer located on the PAC2 board. The transducer is a bridge type transducer which produces a 10mv/PSI signal when excited with a 10 Vdc signal. (Note: The range of the pressure transducer is 0 to 5 psi vacuum.) In addition, there are two ports on the SCX05 pressure transducer. The bellows tube should be connected to the port near the edge of the sensor. The port near the center of the sensor is not used.

An instrumentation amplifier with a fixed gain of 100x converts the differential bridge signal to a single-ended signal. The signal is then sent to a noninverting unity gain amplifier with its inverting input connected to a 12-bit DAC configured as a voltage source (offset control). The output of the DAC varies from 1V to –4V depending on the value written to it by the 80C196 microprocessor. This offset control stage allows for gross pressure offsets developed with the bellows attached to the patient. Up to 4 psi of vacuum and 1 psi pressure can be nulled out with this adjustment. (Note: The transducer is meant to be operated in vacuum mode, so it is important that the bellows <u>not</u> be vented after connection to the patient. The sensor will not be damaged by pressures; however, its output accuracy is guaranteed only for vacuum mode.)

After the offset section, the respiration signal is sent to a 12-bit DAC configured as a digital divider. The divider adjusts the gain of the incoming signal from 1/4096 to 4096/4096 depending on the value that is written to the DAC by the 80C196 microprocessor. Smaller written values correspond to smaller gains. After the adjustable gain stage, the signal is sent to a fixed gain stage set to 200x. The signal is then clipped to +/– 5 volts and sent to a 12-bit A/D converter where the signal is digitized.

5-5 PAC2 Photoplethysmograph Circuit

The photoplethysmograph circuit on the PAC2 board works with either a standard fiber optic type probe, or with a future integrated optical to electrical fiber optic probe.

The standard fiber optic probe is routed in through the front of the PAC2 assembly and is connected to two fiber optic SMA-type connectors that protrude from the PAC2 circuit board enclosure. An eight-pin connector on the front of the PAC2 is provided for the future user replaceable probe (integrated optical to electrical I/F).

Photoplethysmograph Theory

An infrared emitter beams light into the area were blood profusion is to be measured (usually a finger or toe). The light is scattered and/or absorbed in varying amounts relative to the amount of blood profusion in that area. An infrared detector, placed on the skin surface near the emitter, measures the reflected light and produces a corresponding voltage. This voltage is then gained up to useful levels for display.

Immediately following the infrared detector, is a high-pass filter that removes gross offsets and attenuates respiration artifact in the peripheral pulse signal. The resultant signal is amplified by two amplifiers, each with a gain of 20x.

At this point, the signal is sent to an automatic gain control (AGC) circuit which automatically adjusts the signal gain and offset. The circuit attempts to set the output signal level at 3 volts relative to ground. When a patient is connected to the probe, the AGC circuit immediately begins converging on a final gain value. The worst case converge time should be less than one minute, however the signal may reach a useable value in only 10–20 seconds.

A bar graph LED is driven by the analog peripheral pulse signal after the AGC circuit has properly adjusted the signal level. The bar graph lights to ~80% of full scale when the AGC circuit has adjusted the pulse signal to 3 volts.

The peripheral pulse signal is sent from the AGC 12-bit DAC to low-pass and high-pass filters to again eliminate gross offset, and to further reduce pulsation artifacts. The resulting signal is sent to a 10x gain stage and then to clipping network. The clipping network ensures that the signal is never more than a diode drop above 5 volts or below –5 volts. After clipping, the signal is sent to a 12-bit A/D converter where it is digitized.

5-6 Auxiliary Analog Circuits

Two auxiliary analog input ports on the PAC2 board provide an interface for future gating signal inputs. The circuits are unity gain and handle input voltage ranges of +/– 5V relative to nonisolated power ground. The output of the auxiliary circuits is sent directly into the main A/D converter.

6- TEMPERATURE SENSOR IN MAGNET BORE (MG2 A36)

The Bore Wall Temperature Sensor (prior to wide-open enclosure) is installed at the top rear of the magnet bore with a stick-on label, and attaches to the SRI piggyback board. For MR/*i* and CV/*i* wide-open enclosures, the Bore Wall Temperature Sensor is installed in the top of the front endbell. When the bore temperature reaches 29 °C (33 °C for MR/*i* and CV/*i* wide-open enclosures), a message appears on the screen alerting the operator of a warmer than normal condition. The operator can continue to scan after this message has appeared. This message turns off if the bore cools to 27 °C. (30 °C for MR/*i* and CV/*i* wide-open enclosures) At that point, a "returned to normal" message appears.

If the bore wall continues to heat to 33 °C (36 °C for MR/*i* and CV/*i* wide-open enclosures), a second message appears on the screen alerting the operator of a "scan inhibited - patient comfort sensor trip" condition. Once this occurs, the operator is allowed to finish the present scan, but will be unable to start a new scan until the bore temperature has cooled to 31 °C (33 °C for MR/*i* and CV/*i* wide-open enclosures). At this point a "can initiate new scans, but comfort level still warm" bore temperature message appears. Once the bore temperature cools to 27 °C (30 °C for MR/*i* and CV/*i* wide-open enclosures), a "returned to normal" bore temperature message appears.

To lower the magnet bore temperature, check that the patient fan is on, magnet room temperature is below 21 °C, and bore fan air flow is unobstructed.

Note

The programmed temperature level values are located in the MR configuration file.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	May 1, 1998	W. Pasciak	Initial release converted to Word
0	Oct 12, 1999	M. Keber	Added correct proprietary header and wide-open bore temp sensor location and overtemp limits.